

Pricing Theory Assignment #1

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Question 1

(a)

We need to buy stock for $\$K$ at time T . To replicate we can:

1. Borrow $\frac{K}{(1+r)^\tau}$, $\tau = T - t$
2. Buy stock, hold till forward expiry

→ This portfolio is worth

$$\Pi_T = S_T - \frac{K}{(1+r)^\tau} (1+r)^\tau = S_T - K$$

$$V_T = S_T - K \quad (\text{forward contract at time } T)$$

Both our portfolio and the forward have the same value at maturity.

∴ General expression for forward:

$$V_t = S_t - \frac{K}{(1+r)^\tau}$$

(b)

$$V_t = S_t - \frac{K}{(1+r)^\tau} = 55 - \frac{52}{(1.05)^{0.75}} = \$4.87$$

(c)

Sell forward for \$6

Time t		Time T	
Sell forward	+6	Close forward	$52 - S_T$
Buy stock	-55	Share delivered	$+S_T$
Borrow $\frac{K}{(1+r)^\tau}$	+50.13	Return loan	-52
Total	\$1.13	Total	0

Buy forward for \$4

Time t		Time T	
Buy forward	-4	Forward	$S_T - 52$
Sell stock	+55	Close short	$-S_T$
Lend	-50.13	Loan repaid	+52
Total	\$0.87	Total	0

To arb the mispriced forward we use two instruments:

1. Forward V_t
2. Replicating portfolio Π_t

We buy low & sell high, noting that the replicating portfolio always equals the no arb value of the forward.

Question 2

State	0	u	d
Stock (\$)	60	75	48
Put (\$)	?	0	12

$$K = 60, \quad \text{payoff} = (60 - S_T)^+$$

(a)

Let B be the amount of bonds we have. Let N be defined similarly for the stock.

$$\begin{cases} 75N + 1.04B = 0 \\ 48N + 1.04B = 12 \end{cases}$$

which gives

$$27N = -12 \Rightarrow N = -\frac{12}{27} = -\frac{4}{9}$$

Thus

$$1.04B = -75 \left(-\frac{4}{9} \right) \Rightarrow B = 32.05$$

$$V_0 = 60 \left(-\frac{4}{9} \right) + 32.05 = \$5.38$$

(b)

$$1.04q_u + 1.04q_d = 1 \Rightarrow q_u + q_d = \frac{1}{1.04} = \frac{25}{26}$$

$$75q_u + 48q_d = 60 \Rightarrow 75q_u + 48 \left(\frac{25}{26} - q_u \right) = 60$$

$$27q_u = \frac{180}{13} \Rightarrow q_u = \frac{20}{39}, \quad q_d = \frac{35}{78}$$

(c)

$$V_0 = \sum_i V_i q_i = 0 \left(\frac{20}{39} \right) + 12 \left(\frac{35}{78} \right) = \$5.38$$

The feature via FTAP that admits no arb is the fact that the state price vector $\vec{q}$ is strictly positive.

Question 3

New derivative:

$$\delta_0 = 27, \quad \delta_u = 30, \quad \delta_d = 20$$

$$\delta_0 = 30 q_u + 20 q_d = \$24.36 \rightarrow \text{No-arb price}$$

→ With only 2 states, 2 independent assets $\Rightarrow$ 2 state prices.

Thus $\{S, B\}$ spans all possible payoffs in our market.

→ I would not purchase, as it is overpriced.